Day 4: Connecting and Testing Peripherals

Topic: Setting Up and Testing Peripheral Devices

Learning Objectives

After completing this practical session, students will be able to:

- Connect peripheral devices to a computer.
- Install and configure peripheral devices.
- Verify proper device functionality.
- Identify and resolve connectivity issues.
- Test printers, webcams, and other peripherals.

Introduction

Peripheral devices are hardware components connected to a computer to perform specific functions. Proper installation and testing ensure that the devices work correctly and communicate with the computer.

Common Peripheral Devices

- Printer
- Webcam
- Keyboard
- Mouse
- Scanner
- Speakers

- Headphones
- External Storage Devices

Practical Activity 1: Connecting and Testing a Printer

Objective

Connect a printer to a computer and verify that it is functioning properly.

Procedure

Step 1: Connect the printer using a USB cable or network connection.

Step 2: Turn on the printer.

Step 3: Wait for Windows to detect the printer.

Step 4: Open Settings and navigate to Printers and Scanners.

Step 5: Verify that the printer appears in the device list.

Step 6: Set the printer as the default printer.

Step 7: Print a test page.

Step 8: Confirm that the page prints successfully.

Expected Result

The printer is detected by Windows and successfully prints a test page.

Practical Activity 2: Connecting and Testing a Webcam

Objective

Connect a webcam and verify video functionality.

Procedure

Step 1: Connect the webcam to a USB port.

Step 2: Wait for Windows to install the required driver.

Step 3: Open the Camera application.

Step 4: Verify that live video is displayed.

Step 5: Capture a photo or short video.

Step 6: Check image quality and device response.

Expected Result

The webcam is detected and displays live video without errors.

Practical Activity 3: Connecting and Testing Speakers or Headphones

Objective

Verify audio output devices.

Procedure

Step 1: Connect speakers or headphones.

Step 2: Check volume settings.

Step 3: Open a media file.

Step 4: Play audio.

Step 5: Verify clear sound output.

Expected Result

Audio plays correctly through the connected device.

Task: Resolve Connectivity Issues for a Given Setup

Scenario

A computer has multiple peripheral devices connected.
One or more devices are not functioning properly.
Students must identify the cause and resolve the issue.

Troubleshooting Procedure

Step 1: Identify the Device

Determine which peripheral is not working.

Step 2: Check Physical Connections

- Verify USB cables are properly connected.
- Check power cables.
- Inspect ports for damage.

Step 3: Check Device Power

• Ensure the device is powered on.
• Verify indicator lights if available.

Step 4: Verify Device Detection

• Open Device Manager.
• Check whether the device appears in the list.
• Look for warning symbols.

Step 5: Test Different USB Ports

• Disconnect the device.
• Connect it to another USB port.
• Check whether the device is detected.

Step 6: Update or Reinstall Drivers

• Open Device Manager.
• Update device drivers.
• Reinstall drivers if necessary.

Step 7: Test the Device

• Print a test page for a printer.
• Open Camera for a webcam.
• Play audio for speakers.

Step 8: Confirm Resolution

Verify that the peripheral functions correctly after troubleshooting.

Sample Connectivity Issues

Issue 1

Problem:
Printer is connected but does not print.

Possible Solution:
Check printer status, clear print queue, and set as default printer.

Issue 2

Problem:
Webcam shows a black screen.

Possible Solution:
Check camera permissions and reconnect the webcam.

Issue 3

Problem:
USB device is not detected.

Possible Solution:
Try a different USB port and reinstall drivers.

Issue 4

Problem:
No sound from speakers.

Possible Solution:
Check audio output settings and volume levels.

Practical Assessment

Students must:

- Connect a printer and print a test page.
- Connect a webcam and verify live video.
- Identify a connectivity issue provided by the instructor.
- Resolve the issue and demonstrate successful operation.

Outcome

Students will gain hands-on experience in connecting, testing, and troubleshooting common peripheral devices used in daily computer operations.